WORKSHEET

Chapter 10: Mean, variance and standard deviation calculations

1 Find the sample means for each of the following samples of the hourly output on an assembly line.

a 175, 152, 192, 171, 170, 158, 174, 197, 171, 181

b 193, 167, 170, 163, 170, 184, 165, 186, 184, 177

c 174, 176, 161, 170, 189, 187, 173, 169, 185, 178

d 182, 192, 112, 175, 190, 191, 181, 181, 185, 187

© Cengage Learning Australia Pty Ltd 2014

2 Find the sample means for each of the following samples of the number of text messages sent by students in a day.

a 26, 8, 23, 24, 33, 11, 26, 53, 37, 48, 43, 39

b 34, 26, 0, 29, 0, 26, 30, 11, 26, 48, 53, 44

c 66, 6, 25, 28, 18, 41, 25, 19, 8, 41, 5, 14

d 57, 26, 70, 0, 60, 9, 27, 72, 30, 15, 23, 15

3 Calculate the standard deviation and variance (correct to 2 decimal places) for each sample given in Question **1**.

4 Calculate the standard deviation and variance (correct to 2 decimal places) for each sample given in Question **2**.

© Cengage Learning Australia Pty Ltd 2014

5 CAS The average daily global solar exposure (measured in MJ m^{-2}) for an Australian city is about 27 with a standard deviation of 5.5. Simulate 30 samples of 20 daily exposures each and calculate their means.

a Find the mean of the sample means.

b Find the standard deviation of the means.

c Comment on the results.

6 CAS The average weekly amount spent by Australian households on recreation is $160 with a standard deviation of $40. Simulate 20 samples of 40 households each and calculate their means.

a Find the mean of the sample means.

b Find the standard deviation of the means.

c Comment on the results.

© Cengage Learning Australia Pty Ltd 2014

7 CAS The average foot length of Australian men is 27 cm with a standard deviation of 10 cm. Simulate 30 samples of the foot lengths of 25 men each and calculate their means.

a Find the mean of the sample means.

b Find the standard deviation of the means.

c Comment on the results.

8 CAS The average weekly amount of money that tertiary students earn through part time employment is $160 with a standard deviation of $50. Simulate 15 samples of 50 students each and calculate their means.

a Find the mean of the sample means.

b Find the standard deviation of the means.

c Comment on the results.

© Cengage Learning Australia Pty Ltd 2014

Answers

1 **a** 174.1

b 175.9

c 176.2

d 177.6

2 **a** 30.9

b 27.3

c 24.7

d 33.7

3 **a** 12.95; 167.69

b 9.80; 96.09

c 8.35; 69.76

d 22.44; 503.64

4 **a** 13.27; 176.08

b 16.29; 265.35

c 16.97; 288.06

d 23.62; 558.06

5 **a** Results will vary but it will be close to 27.

b It will be close to 1.2.

c The sample mean is close to the mean, but the sample standard deviation is smaller.

6 **a** Results will vary but it will be close to 160.

b It will be close to 7.3.

c The sample mean is close to the mean, but the sample standard deviation is smaller.

7 **a** Results will vary but it will be close to 27 cm.

b It will be close to 1.8.

c The sample mean is close to the mean, but the sample standard deviation is smaller.

8 **a** Results will vary but it will be close to $160.

b It will be close to 8.

c The sample mean is close to the mean, but the sample standard deviation is smaller.

© Cengage Learning Australia Pty Ltd 2014